BEEE Questions (EMP)

Unit-I

DC Circuits -

1. Problems on Voltage division rule and Current division rule?
2. Problem on resistor Simplification?
3. Define ohm's Law. Explain the Limitations of ohms Law?
4. Define KVL and KCL and explain with an example?

Ac circuits:

5. Define and Derive the expressions for RMS value, Average value, Form Factor and Peak Factor? ★
6. What is Power factor. Explain the Impotance / significance of Power factor?
7. Draw the Power triangle. Explain each and every term?
8. Define Amplitude, time period, Frequency, Peak value, Peak to Peak value?

Unit - II

Part - 1

(1) Explain the principle and operation of Single phase transformer?

(2) Explain the principle and operation of DC generator?

(3) Explain the principle and operation of 3 phase Induction motor?

(4) Explain the principle and operation of any two types of Induction motor?
Single phase

(5) Explain the principle and operation of DC Motor?

Part - 2

(1) Explain the operation of wheatstone bridge and derive the expression for unknown resistance?

(2) Explain the operation of Megger with neat diagram?

(3) Explain the principle and operation of PMMC Instrument with a neat diagram?

(4) Explain the principle and operation of Moving Iron repulsion type- Instrument with neat diagram?

(5) Explain the operational difference between Moving coil and Moving Iron type Instruments?

Unit-III

Part 1

(1) Explain the Layout diagram of Thermal Power plant and explain each and every part?

(2) Explain the Layout diagram of Hydro Power plant and explain each and every part?

(3) ~~Draw~~ Explain the Layout diagram of Wind Power plant and explain each and every part?

(4) Explain the Layout diagram of Solar Power plant and explain each and every part?

Part-II

(1) problem on every energy bill?

(2) What is electric shock.
Explain the safety precaution to avoid it?

(3) Explain the operation of Fuse and MCB with neat circuit diagram?

(4) Explain different types of earthings?

Unit-4

Part-I

① Explain the operation of PN junction Diode and Draw VI characteristics?

② Explain the operation of Zenor Diode and Draw VI characteristics?

③ Explain How Zenor Diode acts as a zenor voltage regulator?

Part-II

① Explain the operation of NPN and PNP transistor?

② Explain (CBM) Common base mode of transistor and Draw Input and output Characteristics

③ Explain Common emitter mode of transistor and Draw Input and output Characteristics?

④ Analyse the CB, CE & CC modes of transistors?

Unit-5

Analog electronics

① Explain Half Wave and Full Wave rectifier. Draw the wave forms and derive the expression for output voltage and current?

② Explain the operation of Bridge rectifier and Draw the wave-forms and derive the expression for output voltage and current?

③ Explain the different types of filters and draw the corresponding wave forms?

Digital Electronics:

① Explain different types of Logic gates with Truth Table and Logic diagram?

② Explain Half adder and Full adder. with Logic diagram?

③ Problem on Hamming code?

④ Explain Excess 3 code and Gray code from 0-10.?

⑤ Convertion of Number systems -

Like- $(1101)_{12} = (\quad)_{10}$

$(234)_8 = (\quad)_{10}$

$(2BA5)_{16} = (\quad)_{10}$